IKHIMWIN EMMANUEL

LinkedIn Profile: https://www.linkedin.com/in/emmanuel-ikhimwin-aa311a244

GitHub Portfolio: https://github.com/ikhimwinemmanuel

Portfolio Website: https://ikhimwinemmanuel.github.io/

EDUCATION & CERTIFICATIONS

Master of Artificial Intelligence - Western Sydney University **WAM**: 76

Dean's Merit List Award (Top 10% in the School of Computer, Data & Mathematical Sciences**)**

B.Eng. Electrical & Electronics Engineering - University of Benin

Tableau Desktop Specialist Certified

SKILLS

Programming & Software: Python, SQL, TypeScript, FastAPI, REST APIs, Git/GitHub.
Cloud & DevOps: AWS, Terraform, Docker, AWS Lambda, Amazon ECR, CloudWatch, Linux.
AI & ML: LLMs, AI agents, LangChain, LangGraph, RAG, OpenAI API, Hugging Face**.**

AI Evaluation & Observability: Langfuse, LangSmith, RAGAS, LLM-as-a-judge.

Data & Analytics: Pandas, Snowflake, dbt, Tableau.

Business & Communication: Stakeholder engagement, requirements gathering, technical communication, solution-focused problem solving.

EMPLOYMENT HISTORY

AI & Data Engineering Intern | BILLINI Nov. 2025 – Feb. 2026

- Worked with business stakeholders to understand reporting requirements and built Tableau dashboards covering turnover, headcount, new hires and retention
- Developed and tested an AI analytics agent using Python, LangChain/LangGraph and LLM tool calling, allowing users to query governed Tableau data in natural language.

- Built grounding controls so the agent relied on trusted business data and rejected questions it could not answer reliably.
- Used LangSmith to trace agent behaviour, investigate failures and improve tool-calling and response quality.
- Built API integrations and authentication flow to connect the application securely to Tableau Cloud.

Tableau Dashboards:
https://github.com/ikhimwinemmanuel/Interactive-Dashboard-tableau/blob/main/README.md

FEATURED PROJECTS

MediNotes AI – Healthcare Consultation Assistant

Next.js · TypeScript · FastAPI · OpenAI API · Langfuse · Docker · AWS

GitHub: https://github.com/ikhimwinemmanuel/medinotes-ai

Live Demo: **https://medinotes-ai.vercel.app/**

- Built and deployed a full-stack AI application using Next.js, FastAPI and the OpenAI API.
- Implemented authentication, JWT-protected API access, streaming responses and backend validation using FastAPI and Pydantic.
- Added Langfuse observability to track prompts, responses, latency, token usage, errors and evaluation results.
- Built an LLM-as-a-judge evaluation pipeline using synthetic test cases to identify weak outputs, unsupported claims and failure patterns.
- Containerised the application with Docker and deployed it to AWS Lambda via Amazon ECR and Lambda Web Adapter, with CloudWatch used for application and infrastructure monitoring.

AWS Cloud Web Application

AWS · EC2 · RDS · VPC · Application Load Balancer · Auto Scaling · Secrets Manager · Node.js

GitHub: **https://github.com/ikhimwinemmanuel/cloud-web-app-xyz-university**

- Designed and deployed a cloud web application on AWS using EC2 and RDS within a custom VPC.
- Configured public and private subnets, security groups and networking to separate application and database resources.
- Used an Application Load Balancer and Auto Scaling to improve application availability and scalability.
- Used AWS Secrets Manager to securely manage database credentials and sensitive configuration.
- Deployed and tested the application across the AWS environment and troubleshot connectivity and configuration issues.

AI Analytics Agent

LangChain · LangGraph · Tableau · Python · LLM API · FastAPI · Docker

GitHub: **https://github.com/ikhimwinemmanuel/tableau-langchain-analytics-agent**

Live Demo: **https://ikhimwin-tableau-langchain-analytics-agent.hf.space/ui/**

- Built and deployed an agentic AI application using LangGraph/ReAct to orchestrate LLM tool calls against governed Tableau data.
- Designed grounding and behavioural controls so the agent answered only using retrieved business data and declined unsupported questions.
- Used LangSmith for agent tracing and observability, identifying tool-calling and reasoning failures and improving system behaviour.
- Built a FastAPI backend, secured Tableau access using JWT authentication and containerised the application with Docker.

AI Assistant Chatbot – Retrieval-Augmented Generation

LangChain · ChromaDB · RAG · Hugging Face

GitHub: **https://github.com/ikhimwinemmanuel/ai-assistant-chatbot-rag**

Live Demo: https://huggingface.co/spaces/Ikhimwin/bible-rag-langchain

- Built and evaluated an end-to-end RAG application using dense retrieval, BM25 and Reciprocal Rank Fusion.
- Built an evaluation framework using deterministic tests and RAGAS to measure retrieval quality, faithfulness and answer quality.
- Improved context recall from **0.87 to 0.93** after identifying retrieval failures and introducing hybrid retrieval.
- Added a grounding gate that rejects irrelevant questions before making an LLM call, reducing unsupported responses.